

CASE STUDY

ESG Data Analyzer

End-to-end ESG data pipeline with AI-powered analysis

Python	Claude API	GHG Protocol	Scope 1 / 2 / 3	Excel + pandas
~400 Lines of code Pure Python	1+2+3 Scopes covered 7 categories	30+ Emission factors IPCC, IEA, DEFRA	3 Output files CSV + report + audit log	

Prepared by Christian Seymand ESG Analyst • Data Specialist • AI Workflow Builder B.Sc. Environmental Management, UADE • Sustainability Analyst, ASU	Tech stack Python 3.14 • pandas • openpyxl Anthropic Claude API (claude-sonnet-4-6) Excel • CSV • SQLite
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1. The problem

Companies that need to measure and report their carbon emissions face a data problem before a reporting problem. Information is fragmented, stored in different units, spread across departments, and no one has it centralized. An ESG consultant can spend more time cleaning data than analyzing it.

Without this system	With this system
• Excel files with inconsistent units (MWh, GJ, gallons, kg mixed together) • Missing values with no documented imputation criteria • Manual calculations prone to errors in emission factors • Subjective analysis with no sector benchmarking • Reports that take days to produce • No assumptions log — not audit-ready	• Automatic unit conversion to GHG Protocol standard • Automatic detection and cleaning of missing values using median imputation • Updated emission factors (IPCC AR6, IEA 2023, DEFRA 2023) • AI analysis with ranking, red flags, and SBTi recommendations • Executive report generated in seconds • Assumptions log and data sources included in every output

2. The solution

A Python ESG data pipeline that accepts raw client inputs (Excel with mixed units, missing values, multiple facilities), standardizes them, calculates a complete GHG inventory under the GHG Protocol Corporate Standard, and calls the Claude API to generate an executive analysis with recommendations. All in a single command.

System architecture

Module	What it does	Technology
Unit conversion	MWh→kWh, gallons→litres, GJ→kWh, kg→tonnes	Python dict + functions
Gap detection	Identifies missing values per column and row	pandas isnull()
Imputation	Replaces missing values with dataset median	pandas fillna()
Scope 1 calculation	Combustion + refrigerants × IPCC AR6 factors	Python
Scope 2 calculation	Electricity × country grid factor (IEA)	Python
Scope 3 calculation	Cat 1 spend-based, Cat 4 tonne-km, Cat 6/7 pax-km	Python
AI analysis	Structured prompt → Claude API → executive report	Anthropic SDK
Output	Clean CSV + report TXT + assumptions log	pandas + file I/O
Client template	Excel with formulas, validation, pre-loaded factors	openpyxl

Emission factors used

Category	Source	Coverage
Electricity grid	IEA Emissions Factors 2023	AR, CL, PE, BR, CO, MX
Fossil fuels	IPCC AR6 (2021)	Diesel, gasoline, CNG, LPG, fuel oil
Refrigerants	IPCC AR6 — GWP100	R-410A, R-134a, R-404A
Scope 3 Cat 1	EEIO adapted to LatAm	9 supplier sectors
Scope 3 Cat 4	GLEC Framework v3	Truck, rail, ship, air freight
Scope 3 Cat 6 & 7	DEFRA 2023	9 transport modes

3. The process — from raw data to report

The full workflow from when the client delivers their data to when they receive the executive report takes under 60 seconds of execution time. The consultant's time is reserved exclusively for strategic interpretation and recommendations.

Step	Input	Process	Output
1	Raw client data (Excel / CSV)	Unit conversion to standard	Values in kWh, litres, tonnes
2	Standardized dataset	Missing value detection & cleaning	Complete dataset, no gaps
3	Clean dataset	Scope 1, 2 and 3 calculation	GHG inventory in tCO ₂ e
4	GHG inventory	Claude API call with ESG prompt	AI executive analysis
5	AI analysis	Output file generation	CSV + report + assumptions log

Sample system output

Company ranking by emissions intensity (tCO₂e / USD 1M revenue)
Automatic greenwashing detection: 'GreenTech Ltd has optically green branding but revenue intensity nearly identical to heavy manufacturing'
Red flags: RetailCo Scope 3 at 88.7% with no visible supply chain data
Recommendations: 'Launch SBTi Supplier Engagement program targeting top-20 suppliers'
Documented methodological limitations: EEIO factors are estimates, not primary supplier data

4. Project deliverables

Files delivered	Capabilities demonstrated
- analar_v3.py — complete main script - ESG_Data_Collection_Template.xlsx — client-facing template (6 sheets, cross-referenced formulas) - esg_inventario_completo.csv — clean dataset with calculated indicators - reporte_esg_v3.txt — AI-generated executive analysis - Assumptions log with all emission factors and sources (audit-ready)	- GHG Protocol Corporate Standard — Scope 1, 2 and 3 - Carbon accounting with IPCC, IEA, and DEFRA factors - Data cleaning and validation in Python/pandas - LLM API integration (Anthropic Claude) - Prompt engineering for structured ESG outputs - Advanced Excel with openpyxl (dynamic formulas, cross-sheet references)

Excel template structure

Sheet	Content	Data type
INSTRUCTIONS	Usage guide, color convention, step-by-step	Reference
COMPANY	General data: sector, country, headcount, revenue	Client input
SCOPE_1	Fuels + refrigerants with pre-loaded emission factors	Input + auto-calculation
SCOPE_2	Electricity by facility with country grid factors	Input + auto-calculation
SCOPE_3	Cat 1/4/6/7 with spend-based and distance methodology	Input + auto-calculation
RESULTS	Consolidated GHG inventory + intensity KPIs	Formulas only — do not edit

5. Value delivered

This system is not an academic exercise — it solves a real problem faced by ESG consultancies, companies reporting under IFRS S1/S2, and sustainability teams that need to produce auditable GHG inventories within a reasonable timeframe.

<60 sec Analysis time vs. days manually	~90% Error reduction vs. manual Excel	100% Auditability full assumptions log	N clients Scalability same script
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Direct applications

- ESG consultancies that need to accelerate the GHG inventory data collection and calculation phase
- Companies preparing their first report under IFRS S1/S2 or GRI 305
- Sustainability teams transitioning from manual Excel to a reproducible, audit-ready pipeline
- Upwork projects where the client requests: cleaned dataset + ESG analysis + summary report

Why this is scalable

The script accepts any dataset — change the input data, the pipeline stays the same
Emission factors are stored in dictionaries — updating to new IPCC/IEA versions takes 5 minutes
The Claude prompt is fully parameterizable — adaptable per client or sector
The Excel template can be white-labeled with client branding
Next step: connect with n8n so the client uploads the Excel and receives the report automatically

6. Project roadmap

Phase	What it adds	Technology
v4 — Automation	Client uploads Excel → report delivered by email automatically	n8n / Make.com
v5 — Dashboard	Interactive GHG inventory visualization	Power BI / Streamlit
v6 — Multi-year	Year-over-year emissions comparison + SBTi trajectory tracking	Python + pandas
v7 — RAG	Query ESG frameworks (IFRS S2, GRI 305) in natural language	LangChain + embeddings
v8 — REST API	Endpoint that receives data and returns GHG inventory as JSON	FastAPI + Python

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Python • pandas • Claude API • GHG Protocol • IFRS S1/S2 • GRI • Advanced Excel